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# Import libraries
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, export_text

# Create dataset
data = {
    'Outlook': ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain', 'Rain', 'Overcast',
                'Sunny', 'Sunny', 'Rain', 'Sunny', 'Overcast', 'Overcast', 'Rain'],
    'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool', 'Cool', 'Cool',
                    'Mild', 'Cool', 'Mild', 'Mild', 'Mild', 'Hot', 'Mild'],
    'Humidity': ['High', 'High', 'High', 'High', 'Normal', 'Normal', 'Normal',
                 'High', 'Normal', 'Normal', 'Normal', 'High', 'Normal', 'High'],
    'Wind': ['Weak', 'Strong', 'Weak', 'Weak', 'Weak', 'Strong', 'Strong',
             'Weak', 'Weak', 'Weak', 'Strong', 'Strong', 'Weak', 'Strong'],
    'Play': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes',
             'No', 'Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'No']
}

df = pd.DataFrame(data)

print("Dataset:")
print(df)

# Encode categorical values
le = LabelEncoder()

for column in df.columns:
    df[column] = le.fit_transform(df[column])

# Features and target
X = df.drop('Play', axis=1)
y = df['Play']

# Train ID3 Decision Tree
model = DecisionTreeClassifier(criterion='entropy', random_state=0)
model.fit(X, y)

# Print Decision Tree Rules
print("\nDecision Tree:\n")
print(export_text(model, feature_names=list(X.columns)))

# Predict a new sample
# New Sample: Sunny, Cool, High, Weak
sample = pd.DataFrame({
    'Outlook': ['Sunny'],
    'Temperature': ['Cool'],
    'Humidity': ['High'],
    'Wind': ['Weak']
})

# Encode sample using same mappings
encoders = {}
for column in ['Outlook', 'Temperature', 'Humidity', 'Wind']:
    le = LabelEncoder()
    le.fit(data[column])
    encoders[column] = le
    sample[column] = le.transform(sample[column])

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prediction = model.predict(sample)

# Decode output
play_encoder = LabelEncoder()
play_encoder.fit(data['Play'])

print("\nPrediction:")
print("Play Tennis =", play_encoder.inverse_transform(prediction)[0])

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Dataset:

	Outlook	Temperature	Humidity	Wind	Play
0	Sunny	Hot	High	Weak	No
1	Sunny	Hot	High	Strong	No
2	Overcast	Hot	High	Weak	Yes
3	Rain	Mild	High	Weak	Yes
4	Rain	Cool	Normal	Weak	Yes
5	Rain	Cool	Normal	Strong	No
6	Overcast	Cool	Normal	Strong	Yes
7	Sunny	Mild	High	Weak	No
8	Sunny	Cool	Normal	Weak	Yes
9	Rain	Mild	Normal	Weak	Yes
10	Sunny	Mild	Normal	Strong	Yes
11	Overcast	Mild	High	Strong	Yes
12	Overcast	Hot	Normal	Weak	Yes
13	Rain	Mild	High	Strong	No

Decision Tree:

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|--- Outlook <= 0.50
|   |--- class: 1
|--- Outlook > 0.50
|   |--- Humidity <= 0.50
|   |   |--- Outlook <= 1.50
|   |   |   |--- Wind <= 0.50
|   |   |   |   |--- class: 0
|   |   |   |   |--- Wind > 0.50
|   |   |   |   |--- class: 1
|   |   |--- Outlook > 1.50
|   |   |   |--- class: 0
|   |--- Humidity > 0.50
|   |   |--- Wind <= 0.50
|   |   |   |--- Temperature <= 1.00
|   |   |   |   |--- class: 0
|   |   |   |   |--- Temperature > 1.00
|   |   |   |   |--- class: 1
|   |   |--- Wind > 0.50
|   |   |   |--- class: 1

```

Prediction:

Play Tennis = No

